



VECTORIZED SIGNAL STAR

VSS-1

16-BIT SIMD CORE

EEE4120F · 2026

# A SIMD-lite extension of StarCore-1 — twice the byte throughput, one instruction.

*"For when every cycle on orbit counts."*

Satellites generate floods of byte-packed sensor data — temperature channels, voltage rails, GPIO status. The baseline StarCore-1 forces you to **unpack, compute, repack** every word. **VSS does both bytes at once**, with a single new opcode and a lane-split ALU. Same pipeline. Same toolchain. Half the instruction count on the workloads that hurt most.

## WHAT IT DOES



### Dual 8-bit lanes

The ALU splits the carry chain at bit 7. Two independent byte ops, one clock.



### One new opcode

4'b1110 (VECTOR). 8 vector ops via funct[2:0]. Reserved 1010 untouched.



### Zero scalar regression

vector\_mode = 0 keeps every Practical 4 instruction bit-for-bit identical.



### Drop-in datapath

Two new wires (vector\_mode, funct), one new ALUOp encoding. That's the whole edit.

2×

BYTES / INSTR

6×

MODELLED SPEEDUP\*

39/39

GOLDEN TESTS PASS

## SPEC SHEET

|                 |                                        |
|-----------------|----------------------------------------|
| ARCHITECTURE    | 16-bit single-cycle RISC, Harvard      |
| VECTOR ISA      | VADD VSUB VINV VSHL VSHR VAND VOR VSLT |
| ENCODING        | opcode 4'b1110, funct = instr[2:0]     |
| MODES           | scalar 16b / SIMD-lite 2 × 8b          |
| MODULES TOUCHED | ALU.v · ALU_Control.v · ControlUnit.v  |
| VERIFIED ON     | Icarus Verilog 12.0 + GTKWave          |